HDOC- 3 – 1A

NAME : DARSHIL RAJ

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Program to Convert Temperature from Celsius to Fahrenheit

This document provides a clean, well-commented implementation for converting temperature from Celsius to Fahrenheit in C. The program takes a temperature value in Celsius as input, applies the standard conversion formula, and displays the Fahrenheit temperature.

1. C Implementation

Step 1: Understand the Mathematical Formula

• Fahrenheit = (Celsius × 9/5) + 32

Step 2: Define Variables and Data Types

• Use a floating-point type (double) to store Celsius and Fahrenheit temperatures. This allows the program to correctly handle both decimal values and integer inputs.

Step 3: Take User Input

• Read the Celsius temperature using scanf("%lf", &celsius;).

Step 4: Perform the Calculation

1. Compute fahrenheit = (celsius × 9/5) + 32.

Step 5: Format the Output

• Print the converted Fahrenheit temperature using formatted output specifications.

Complete C Code

```c
/*
 * Q3 (User Inputs, Operations & Output)
 * Write a program to convert temperature from Celsius to Fahrenheit.
 */

#include <stdio.h>

int main() {
    double celsius, fahrenheit;

    // Step 1: Read input value for Celsius temperature
    if (scanf("%lf", &celsius) == 1) {

        // Step 2: Convert Celsius to Fahrenheit
        fahrenheit = (celsius * 9.0 / 5.0) + 32.0;

        // Step 3: Print output in required format
        printf("Fahrenheit=%.2f\n", fahrenheit);
    }

    return 0;
}
```

Dry Run with Test Cases

Sample Test Case 1:

• Input: 25

• Calculation:

o Fahrenheit = (25 × 9/5) + 32 = 77

• Output: Fahrenheit=77.00

Sample Test Case 2:

• Input: 0

• Calculation:

o Fahrenheit = (0 × 9/5) + 32 = 32

• Output: Fahrenheit=32.00